

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA0713
NETWORKS

COURSE NAME: COMPUTER

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I LOKESH S (192572151) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S LOKESH

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X^3+1 .

- i.) Use polynomial long division to determine the message that should be transmitted.
- ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8)** error-correcting code.

4. a) A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

b) A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also,

calculate the total number of possible host addresses available within this network based on the given prefix length.

5. A university is conducting an online semester examination for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps high-speed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps)**, **home Wi-Fi (20 Mbps)**, and **mobile networks (5–10 Mbps)**. During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. **Analyze** the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X^3+1 .

- i.) Use polynomial long division to determine the message that should be transmitted.**
- ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?**

Given:

- **Message (Data):** 11100011
- **Generator Polynomial:** $G(x) = x^3 + 1$

Step 1: Convert the polynomial to binary

The polynomial is:

$$G(x) = x^3 + 1$$

Binary representation:

- $x^3 = 1$
- $x^2 = 0$
- $x^1 = 0$
- $x^0 = 1$

So,

Generator = 1001

Degree of polynomial = 3, so append **3 zeros** to the message.

Original message:

11100011

Append 3 zeros:

11100011000

(i) Polynomial Long Division

Divide:

11100011000 $\div$ 1001

Perform XOR division.

1110 $\oplus$ 1001 = 0111

Bring down next bit $\rightarrow$ 1110

1110 $\oplus$ 1001 = 0111

Bring down next bit $\rightarrow$ 1111

1111 $\oplus$ 1001 = 0110

Bring down next bit $\rightarrow$ 1100

1100 $\oplus$ 1001 = 0101

Bring down next bit $\rightarrow$ 1010

1010 $\oplus$ 1001 = 0011

Bring down next bit $\rightarrow$ 0110

(Leading bit is 0, so no XOR)

Bring down next bit → 1100

$$1100 \oplus 1001 = 0101$$

The final **remainder** = **101**

Transmitted Message

Append the remainder to the original message.

Original data:

11100011

CRC: 101

Transmitted message 11100011101

(ii) Leftmost Bit is Inverted

Original transmitted message

11100011101

After the first bit is flipped

01100011101

The receiver divides this received bit stream by the same generator (1001).

Performing CRC division gives the remainder

110

Result

Since the remainder is

$110 \neq 000$

the received message is **not divisible** by the generator polynomial.

Therefore, the receiver concludes that **an error has occurred during transmission**.

Final Answer

(i)

- Generator Polynomial: **1001**
- Message after appending zeros: **11100011000**
- CRC Remainder: **101**
- **Transmitted Message:** 11100011101

(ii)

After the leftmost bit is inverted:

Received Message = 01100011101

CRC calculation gives:

Remainder = 110

Since the remainder is **not 000**, the receiver detects that the data has been corrupted.

Error Detection

The receiver knows an error has occurred because the received codeword is **not exactly divisible by the generator polynomial (1001)**. A non-zero remainder indicates a transmission error.

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

(a) Throughput of the Network

Given:

- Bandwidth = **10 Mbps**
- Number of frames = **12,000 frames/minute**
- Size of each frame = **10,000 bits**

Step 1: Calculate total bits transmitted per minute

$$\begin{aligned}\text{Total bits} &= 12,000 \times 10,000 \\ &= 120,000,000 \text{ bits/minute}\end{aligned}$$

Step 2: Convert to bits per second

$$\begin{aligned}\text{Throughput} &= \frac{120,000,000}{60} \\ &= 2,000,000 \text{ bps} \\ &= 2 \text{ Mbps}\end{aligned}$$

Answer:

Throughput = 2 Mbps

Efficiency of the network:

$$\begin{aligned}\text{Efficiency} &= \frac{2}{10} \times 100 \\ &= 20\%\end{aligned}$$

(b) Channel Capacity

Given:

- Bandwidth $B = 3000\text{Hz}$
- Signal-to-Noise Ratio (SNR) = 3162

Use **Shannon Capacity Formula:**

$$C = B \log_2(1 + SNR)$$

Substitute values:

$$\begin{aligned}C &= 3000 \log_2(1 + 3162) \\ C &= 3000 \log_2(3163)\end{aligned}$$

Using:

$$\begin{aligned}\log_2(3163) &= \frac{\log_{10}(3163)}{\log_{10}(2)} \\ &= \frac{3.5}{0.301} \\ &\approx 11.63\end{aligned}$$

Therefore,

$$\begin{aligned}C &= 3000 \times 11.63 \\ &= 34,890 \text{ bps} \\ &\approx 34.9 \text{ kbps}\end{aligned}$$

Answer:

$$\boxed{\text{Channel Capacity} \approx 34.9 \text{ kbps}}$$

Final Answers:

- (a) Throughput = $\boxed{2 \text{ Mbps}}$
- (b) Channel Capacity = $\boxed{34.9 \text{ kbps}}$

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the Hamming (12,8) error-correcting code.

Given

A multispecialty hospital uses wireless patient-monitoring devices to transmit patients' heart rate and blood pressure readings to a central monitoring station. During wireless transmission, **one bit may be corrupted** due to electromagnetic interference. To detect and correct this error **without retransmission**, the hospital uses the **Hamming (12,8) code**.

Step 1: Data to be transmitted

Assume the 8-bit patient data is:

10110011

Step 2: Place parity bits

In Hamming (12,8), there are:

- **8 data bits**
- **4 parity bits (P1, P2, P4, P8)**

Parity bits are placed at positions:

Position	1	2	3	4	5	6	7	8	9	10	11	12
Bit	P1	P2	D1	P4	D2	D3	D4	P8	D5	D6	D7	D8

Insert data bits (10110011):

Position	1	2	3	4	5	6	7	8	9	10	11	12
Value	P1	P2	1	P4	0	1	1	P8	0	0	1	1

Step 3: Calculate parity bits (Even Parity)

P1 checks positions:

1, 3, 5, 7, 9, 11

Values:

P1, 1, 0, 1, 0, 1

Number of 1s = 3 (odd)

So,

P1 = 1

P2 checks positions:

2, 3, 6, 7, 10, 11

Values:

P2, 1, 1, 1, 0, 1

Number of 1s = 4 (even)

So,

P2 = 0

P4 checks positions:

4, 5, 6, 7, 12

Values:

P4, 0, 1, 1, 1

Number of 1s = 3 (odd)

So,

P4 = 1

P8 checks positions:

8, 9, 10, 11, 12

Values:

P8, 0, 0, 1, 1

Number of 1s = 2 (even)

So,

P8 = 0

Step 4: Final transmitted code

Position 1 2 3 4 5 6 7 8 9 10 11 12

Bit 1 0 1 1 0 1 1 0 0 0 1 1

Transmitted Codeword:

101101100011

Step 5: Assume one-bit error

Suppose **bit 7** changes during transmission.

Received code:

101101000011

Step 6: Receiver checks parity

The receiver recalculates the parity bits and forms the **syndrome**:

- S1 = 1
- S2 = 1
- S4 = 1

- $S_8 = 0$

Syndrome:

$0111_2 = 7$

This indicates that **bit 7** is incorrect.

Step 7: Correct the error

Flip bit 7 back to its original value.

Corrected code:

101101100011

Extract the data bits:

10110011

The original patient data is recovered successfully.

Final Answer

- **Data bits:** 10110011
- **Hamming (12,8) codeword:** 101101100011
- **Error introduced:** Bit 7 flipped
- **Syndrome:** $0111_2 = 7$
- **Error detected at:** Position 7
- **Corrected data:** 10110011

4. a) A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

Given

- Network Address: 172.16.10.0/24
- Subnet Mask: /27

Step 1: Calculate the Number of Subnets

Original prefix = /24

New prefix = /27

Number of borrowed bits:

$$27 - 24 = 3$$

Number of subnets:

$$2^3 = 8$$

Answer:

Total Subnets = 8

Step 2: Calculate the Number of Usable Hosts

Remaining host bits:

$$32 - 27 = 5$$

Total addresses per subnet:

$$2^5 = 32$$

Usable hosts:

$$32 - 2 = 30$$

(2 addresses are reserved for Network and Broadcast.)

Answer:

Usable Hosts per Subnet = 30

Step 3: List the /27 Subnets

Each subnet increases by 32.

Subnet Network Address Host Range Broadcast Address

1	172.16.10.0	1 – 30	31
2	172.16.10.32	33 – 62	63
3	172.16.10.64	65 – 94	95
4	172.16.10.96	97 – 126	127
5	172.16.10.128	129 – 158	159
6	172.16.10.160	161 – 190	191
7	172.16.10.192	193 – 222	223
8	172.16.10.224	225 – 254	255

Step 4: Determine the Subnet for IP Address 172.16.10.78

The IP 172.16.10.78 falls within:

- Network Address = 172.16.10.64
- Host Range = 172.16.10.65 – 172.16.10.94
- Broadcast Address = 172.16.10.95

Answer:

- Subnet: 172.16.10.64/27
- Network Address: 172.16.10.64
- Broadcast Address: 172.16.10.95

Step 5: Check Whether 172.16.10.95 is in the Same Subnet

For subnet 172.16.10.64/27:

- Network Address = 172.16.10.64
- Host Range = 172.16.10.65 – 172.16.10.94
- Broadcast Address = 172.16.10.95

The IP 172.16.10.95 is the broadcast address, not a usable host address.

Answer:

- Yes, 172.16.10.95 belongs to the same subnet, but it is the broadcast address.
- Therefore, it cannot be assigned to a host/device.

Final Answers

1. Total Subnets:

8

2. Usable Hosts per Subnet:

30

3. Subnet containing 172.16.10.78:

172.16.10.64/27

4. Network Address: 172.16.10.64
5. Broadcast Address: 172.16.10.95
6. Does 172.16.10.95 belong to the same subnet?
 - Yes, it is in the same subnet.
 - It is the broadcast address, so it is not a usable host IP.

b) A network uses the Internet Protocol version 6 address

2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Given

IPv6 Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 1: Compress the IPv6 Address

Rules for IPv6 compression:

- Remove leading zeros in each block.
- Replace one continuous sequence of all-zero blocks with :: (only once).

Original:

2001:0db8:0000:0000:0000:ff00:0042:8329

After removing leading zeros:

2001:db8:0:0:0:ff00:42:8329

Replace the consecutive zero blocks with ::

2001:db8::ff00:42:8329

Compressed Address

2001:db8::ff00:42:8329

Step 2: Expand the Compressed Address

Expand :: back to three groups of 0000 and restore leading zeros:

2001:0db8:0000:0000:0000:ff00:0042:8329

Expanded Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Determine the Network Prefix (/64)

A /64 prefix means:

- First 64 bits (first 4 blocks) represent the network prefix.
- Last 64 bits (last 4 blocks) represent the interface (host) ID.

Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Network Prefix

2001:0db8:0000:0000::/64

or in compressed form:

2001:db8::/64

Answer

$$\text{Network Prefix} = 2001:\text{db8}::/64$$

Step 4: Calculate the Number of Host Addresses

With a /64 prefix:

- Total IPv6 bits = 128
- Network bits = 64
- Host bits = 64

Number of possible host addresses:

$$2^{64} \\ = 18,446,744,073,709,551,616$$

Approximately:

$$1.84 \times 10^{19} \text{ host addresses}$$

Final Answers

1. Compressed IPv6 Address:
 - 2001:db8::ff00:42:8329
2. Expanded IPv6 Address:
 - 2001:0db8:0000:0000:0000:ff00:0042:8329
3. Network Prefix (/64):
 - 2001:db8::/64 (or 2001:0db8:0000:0000::/64)
4. Total Possible Host Addresses:

$$2^{64} = 18,446,744,073,709,551,616 \text{ host addresses}$$

5. A university is conducting an online semester examination for 8,000 students simultaneously. At exactly 10:00 AM, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a 10 Gbps high-speed network, while students access the examination portal using different internet connections such as fiber broadband (100 Mbps), home Wi-Fi (20 Mbps), and mobile networks (5–10 Mbps). During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. Analyze the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

Primary Networking Problem

The primary networking problem is Flow Control.

Flow control is a mechanism that ensures the sender transmits data at a rate that the receiver can receive and process. In this scenario, the university server sends encrypted question papers at a very high speed, while many students have slower internet connections and devices that cannot process the incoming packets quickly.

Effect of Sender–Receiver Speed Mismatch

- The university server is connected through a 10 Gbps network and transmits data at a very high rate.
- Students use different internet connections:
 - Fiber Broadband: 100 Mbps
 - Home Wi-Fi: 20 Mbps
 - Mobile Network: 5–10 Mbps
- Students with slower connections cannot receive packets as fast as the server sends them.
- The incoming packets accumulate in the receiver's buffer (temporary memory).
- When the buffer becomes full, buffer overflow occurs and packets are discarded.
- Since some packets are lost, the receiver cannot acknowledge them.
- The sender assumes the packets were lost and retransmits them.
- These repeated retransmissions increase network traffic, causing congestion and further delays.
- As a result, students experience:
 - Incomplete downloads
 - Delayed acknowledgments
 - Temporary freezing of the examination portal
 - Slow delivery of question papers

Role of Flow Control

Flow control prevents these problems by regulating the sender's transmission speed according to the receiver's capacity.

In TCP, mechanisms such as the Sliding Window Protocol allow the receiver to advertise how much data it can accept. The sender adjusts its transmission rate based on this information, preventing receiver buffer overflow and reducing unnecessary retransmissions.

Conclusion

The issue arises because the 10 Gbps examination server transmits data much faster than many students' devices and internet connections can receive and process. This mismatch causes receiver buffer overflow, packet loss, retransmissions, increased network traffic, and delays. Implementing flow control ensures that data is sent at a rate the receiver can handle, enabling reliable and efficient delivery of the examination papers.